

Software Test Plan (STP) - Automated Engineering Admission & Counselling Management Platform

Project: Automated Engineering Admission & Counselling Management Platform for KCET and COMEDK (Team 5)

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1. Introduction

Purpose: This document defines the test plan for the Automated Engineering Admission & Counselling Management Platform v1.0. It outlines objectives, scope, strategy, resources, schedule, and responsibilities for testing, and is derived from the Software Requirements Specification (SRS) prepared by Team 5.

Scope: Testing covers the 20 functional requirements (ADM-F-001 to ADM-F-020), 5 non-functional requirements (ADM-NFR-001 to ADM-NFR-005) and 5 security requirements (SEC-F-001 to SEC-F-005) of the SRS: candidate registration and login, profile and exam details, application drafting and submission, document upload and verification, eligibility verification, college/branch/seat-matrix management, choice filling and locking, rank-and-preference-based seat allotment, result publication, seat accept/reject decision, and administrative dashboard and audit logging. Integration with real KEA/COMEDK systems and the other items listed in Section 4 are excluded.

References: SRS - Automated Engineering Admission & Counselling Management Platform for KCET and COMEDK (Team 5), including its Section 6 workflow acceptance tests, Section 7 use-case diagrams and Section 8 Requirements Traceability Matrix; Test Plan Template for SE; IEEE 829 / ISO/IEC/IEEE 29119 (test documentation guidance).

Definitions: KCET (Karnataka Common Entrance Test), COMEDK (Consortium of Medical, Engineering and Dental Colleges of Karnataka), KEA (Karnataka Examinations Authority), SRS (Software Requirements Specification), RTM (Requirements Traceability Matrix), NFR (Non-Functional Requirement), RBAC (Role-Based Access Control), Seat Matrix (record of total, filled and available seats per college, branch and category), Allotment Run (one execution of the seat allotment engine), TC (Test Case), UAT (User Acceptance Testing).

2. Test Items

- Account Module (registration, login/logout, profile and exam details, authentication, RBAC, password storage)
- Application Module (draft save, edit, submission, reference-number generation)
- Document Module (upload and validation, admin verification, status view, resubmission)

- Eligibility Module (rule-based eligibility status and reason)
- Catalogue Module (college/branch catalogue, seat matrix, admin-managed configuration)
- Choice Module (add/remove/preview, reorder, lock)
- Allotment Module (rank + preference + seat-availability engine, Not Allocated handling, seat-count consistency)
- Result Module (result publication, result viewing, accept/reject decision)
- Admin Module (candidate, application and document-verification dashboard)
- Audit Module (audit log of privileged actions)
- Cross-cutting concerns (response time, usability, HTTPS transport security)

3. Features to be Tested

Features mapped to SRS requirement IDs:

- ADM-F-001: Candidate registration (duplicate email/mobile rejected)
- ADM-F-002: Candidate login/logout
- ADM-F-003: Profile view/update and exam-detail entry
- ADM-F-004: Save application as draft
- ADM-F-005: Edit and submit application; unique reference number
- ADM-F-006: Document upload with file type (PDF/JPEG/PNG) and size validation
- ADM-F-007: Admin document verification (Approve/Reject with reason)
- ADM-F-008: Document status view and resubmission of rejected documents
- ADM-F-009: Eligibility evaluation (Eligible / Not Eligible / Pending Verification) with reason
- ADM-F-010: College/branch catalogue with seat availability and filter
- ADM-F-011: Admin college/branch/category-wise seat-matrix management
- ADM-F-012: Choice add/remove/preview with duplicate prevention
- ADM-F-013: Choice reordering
- ADM-F-014: Choice locking
- ADM-F-015: Rank/preference/seat-availability based allotment engine
- ADM-F-016: Not Allocated determination
- ADM-F-017: Admin publication of allotment results
- ADM-F-018: Candidate allotment result viewing
- ADM-F-019: Seat accept or reject/withdraw (exactly once)
- ADM-F-020: Admin consolidated dashboard
- ADM-NFR-001: Response time ≤ 2 s for standard candidate requests
- ADM-NFR-002: Allotment run completes within agreed limit (under 1 minute) on the demonstration dataset
- ADM-NFR-003: Seat counts consistent with allotment records; no allocation exceeds capacity
- ADM-NFR-004: First-time user can complete application submission unaided
- ADM-NFR-005: College/branch/seat/eligibility data configurable via admin interface only
- SEC-F-001: Mandatory authentication
- SEC-F-002: Role-based access control
- SEC-F-003: Salted password hashing

- SEC-F-004: Encrypted transport (HTTPS/TLS)
- SEC-F-005: Audit logging of privileged actions

4. Features Not to be Tested

The following are explicitly out of scope in the SRS and will not be tested:

- Integration with real KEA/COMEDK systems or APIs (all exam and eligibility data is mock/seeded)
- OCR or AI-based document verification (verification is manual by the administrator)
- Aadhaar or biometric authentication; offline/kiosk modes
- Real or simulated payment gateway and SMS gateway; mandatory external email services
- Medical/dental admissions and post-admission academic management
- In-app notifications
- Multiple or subsequent counselling rounds (only a single allotment cycle is implemented)
- AI-based prediction, recommendation or graph-analytics functionality
- Behaviour at real-world admission volumes (the system is validated only for tens to a few hundred candidate records)
- Internals of third-party frameworks, the database engine and the web server (assumed tested by their vendors)

5. Test Approach / Strategy

Levels:

- Unit tests (module-level: eligibility rules, allotment engine, validation logic, password hashing)
- Integration tests (UI ↔ backend ↔ database; module hand-offs such as Document → Eligibility → Choice → Allotment → Result)
- System tests (end-to-end workflows from registration through seat decision, for both Candidate and Administrator roles)
- Acceptance tests (UAT against the workflow acceptance criteria in SRS Section 6)

Types:

- Functional testing (all ADM-F requirements)
- Regression testing (re-run of the automated suite after each defect fix and before each release candidate; the allotment oracle test is mandatory in every regression run)
- Performance testing (response time, allotment run time)
- Usability testing (unassisted first-time-user walkthrough, responsive layout on desktop and mobile browsers)
- Data-consistency testing (seat counts vs. allotment records)
- Security testing (see Section 5.1)

Test design techniques:

- Equivalence partitioning and boundary value analysis (file type and size limits, mandatory fields)
- Decision tables (eligibility outcome by exam type, rank, category and document status)
- State-transition testing (application: Draft → Submitted; document: Pending Verification →

Approved/Rejected → Pending Verification; choice list: Open → Locked; result: Unpublished → Published; seat decision: Undecided → Accepted/Rejected)

- Oracle-based testing for the allotment engine: a small seeded dataset with a hand-computed expected result (Appendix B) plus a larger dataset for scale checks

Defect severity: Critical (blocks a core workflow or corrupts allotment/seat data), High (major feature fails, no workaround), Medium (feature fails, workaround exists), Low (cosmetic or minor).

Entry Criteria: Stable build deployed to the test environment, smoke test passed, seeded test data loaded (candidates, exam records, eligibility rules, colleges, seat matrix), HTTPS configured, SRS baselined.

Exit Criteria: 100% of planned test cases (TC-001 to TC-030) executed, 0 Critical and 0 High defects open, remaining defects triaged with agreed disposition, all SRS acceptance criteria satisfied, 100% requirement coverage in the RTM, and the allotment oracle test passes.

5.1 Security Validation

Mandatory (SRS security requirements):

- Unauthenticated access to every protected page and endpoint is redirected to login with no data returned (TC-021)
- RBAC: candidate accounts denied all administrator functions - seat-matrix edit, allotment run, result publication (TC-022)
- Password storage uses salted cryptographic hashes; no plaintext in the credential store or application logs (TC-023)
- All traffic uses HTTPS/TLS 1.2+; traffic capture shows no plaintext credentials or personal data (TC-024)
- Audit entries (actor, timestamp, action type) exist for every privileged action and cannot be edited via the UI (TC-025)

Additional checks (recommended to protect SRS Security Objectives 1-3):

- Horizontal access control: a candidate cannot view or modify another candidate's profile, application, documents, choices or result by changing IDs in URLs/requests
- Input validation and injection checks (SQL injection, XSS) on registration, profile, application and filter fields
- File upload hardening: renamed or spoofed files (e.g., .exe renamed to .pdf) are rejected; uploaded files are not executable from the server
- Session handling: logout invalidates the session; session cannot be reused after logout
- Admin-only API endpoints return an authorization error when called directly (not only hidden in the UI)

6. Test Environment

Hardware: Developer/test laptops or a college-lab server hosting the application and database; desktop PCs and at least one mobile device (or browser device emulation) as clients. No specialised hardware, kiosk terminals or government network access is required

(SRS 2.4, 3.2).

Software: Platform build v1.0 on the team's chosen web application server and relational/document database (technology stack is not mandated by the SRS); HTTPS enabled with a TLS certificate (a self-signed or test certificate is acceptable); browsers - Chrome, Firefox and Edge (latest versions), in desktop and mobile viewports. No external KEA, COMEDK, payment or SMS services.

Tools: Selenium or Playwright (UI automation), Postman (API and RBAC checks), JMeter (response-time and allotment-scale tests), unit-test framework of the chosen stack (e.g., JUnit/PyTest/Jest), OWASP ZAP and browser developer tools (security checks), Wireshark or equivalent (TLS traffic inspection), SQL client (database/seat-count verification), Jira or GitHub Issues (defect tracking).

Test Data: Mock/seeded data only - sample candidates and exam records (KCET/COMEDK type, rank, category, score); eligibility rule set; colleges, branches and category-wise seat matrix; sample PDF/JPEG/PNG documents plus invalid files (wrong type, oversized); separate admin and candidate accounts; the small allotment oracle dataset (Appendix B) and a larger dataset of about 300 candidates for scalability checks.

7. Test Schedule

Milestones (proposed; to be confirmed against the project calendar):

- Test plan review and baseline: 06-Oct-2026
- Test case design: 08-Oct-2026
- Environment setup and test data seeding: 12-Oct-2026
- Test execution start: 14-Oct-2026
- Test execution end (including regression): 28-Oct-2026
- UAT: 30-Oct-2026 to 02-Nov-2026
- Test Summary Report: 03-Nov-2026

8. Test Deliverables

- Test Plan (this document)
- Test Cases (manual & automated) - TC-001 to TC-030, Appendix A
- Test Scripts (Selenium/Playwright, Postman collections, JMeter plans, SQL consistency checks)
- Test Data (seed scripts, allotment oracle dataset with expected results, sample documents)
- Test Execution Logs
- Defect Reports
- Updated RTM with execution status
- Test Summary Report

9. Roles and Responsibilities

Role	Name	Responsibility
QA Lead		Prepare plan, coordinate execution, maintain RTM, prepare Test Summary Report
Test Engineer		Design & execute test cases, automate regression suite, log defects
Developer		Seed test data, support defect fixes and triage, maintain test environment
Product Owner	Project Guide / Faculty Evaluator	Approve test results, sign-off readiness

Note: Role assignment is proposed for this four-member team; roles may be rotated or combined as needed.

10. Risks and Mitigation

Risk	Mitigation
Delay in stable build delivery	Request early smoke builds from the development team; test module-by-module as each becomes available
Test environment downtime (shared lab server or laptop failure)	Maintain a backup environment on a cloud VM or second laptop; keep seed scripts to rebuild quickly
Allotment engine is the core of the system and hard to verify manually	Use the hand-computed oracle dataset (Appendix B); automate it in every regression run; independent review of expected results by a second tester
SRS leaves some values unspecified (maximum file size, eligibility rules, mandatory fields, behaviour on exam-detail mismatch)	Agree concrete values in a test-data note before test case design; record them as assumptions (Section 11)
Small four-member team with a single-semester timeline	Prioritise High-priority requirements first; automate repeat checks; run security and performance tests in parallel
Seeded data errors cause false failures	Review and freeze seed data before execution; version-control seed scripts
Time-consuming manual usability sessions	Fix a small participant group (about 5 users) and a script in advance

11. Assumptions & Dependencies

- All exam ranks, categories and eligibility rules are mock/seeded and do not represent real candidates or KEA/COMEDK data (SRS assumption)
- A demonstration dataset of tens to a few hundred candidates is sufficient to validate the allotment engine (SRS assumption)
- Document verification is manual by the administrator; no OCR/AI verification is tested
- A single allotment cycle is tested; multiple counselling rounds are not
- No payment functionality is tested
- The maximum upload file size, the eligibility rule set and the list of mandatory application fields will be fixed by the team before test case design (the SRS does not state them); the plan assumes a 2 MB limit for boundary testing
- The system behaviour when entered exam details do not match any seeded record is to be confirmed with the requirements owner; TC-003 will be updated accordingly
- TC-001 to TC-025 take their acceptance criteria directly from the SRS; TC-026 to TC-030 are defined in the SRS RTM by ID only, and their scenarios and thresholds in this plan (e.g., 25 concurrent users) are proposals for review
- The test environment, seed data and a stable build will be available before execution begins
- The development team will fix defects within the agreed triage turnaround so regression can be completed before UAT

12. Suspension & Resumption Criteria

Suspend testing if:

- Environment unavailable for >4 hours
- Build is too unstable (blocks >30% test cases)
- A Critical defect in login, application submission, or the allotment engine blocks dependent test cases
- Seed data is found to be corrupt or inconsistent with the seat matrix

Resume testing if:

- Blocking defects are resolved and verified by a smoke test
- Environment stabilized
- Seed data restored and re-validated

13. Test Case Management & Traceability

RTM ensures mapping of SRS requirements to test cases. Test case IDs follow the SRS (TC-001 to TC-030). Every functional, non-functional and security requirement has at least one test case, so requirement coverage is 30 of 30 (100%) at plan time. Execution status is tracked in the RTM using the SRS status values (Planned / Implemented / Tested / Verified); all entries are currently Planned.

Example:

- ADM-F-001 (Candidate registration) → TC-001
- ADM-F-015 (Allotment engine) → TC-015 (oracle dataset, Appendix B)

- ADM-NFR-001 (Response time) → TC-026

- SEC-F-002 (RBAC) → TC-022

The full matrix follows; detailed test cases are in Appendix A.

13.1 Requirements Traceability Matrix

Req. ID	Summary	SRS §	Test Case	Test Level / Type
ADM-F-001	Candidate registration	4.1	TC-001	Functional, negative, system
ADM-F-002	Candidate login/logout	4.1	TC-002	Functional, negative, system
ADM-F-003	Candidate profile & exam-detail management	4.1	TC-003	Functional, integration
ADM-F-004	Save application as draft	4.2	TC-004	Functional, state transition
ADM-F-005	Edit, submit application; generate reference number	4.2	TC-005	Functional, negative, state transition
ADM-F-006	Document upload and validation	4.3	TC-006	Functional, boundary, negative
ADM-F-007	Document verification (approve/reject)	4.3	TC-007	Functional, role-based
ADM-F-008	Document status view and resubmission	4.3	TC-008	Functional, state transition
ADM-F-009	Eligibility calculation and reason display	4.4	TC-009	Functional, decision table, integration
ADM-F-010	College/branch catalogue view	4.5	TC-010	Functional, data consistency
ADM-F-011	Admin college/branch/seat management	4.5	TC-011	Functional, integration
ADM-F-012	Choice add/remove/preview	4.6	TC-012	Functional, negative
ADM-F-013	Choice reordering	4.6	TC-013	Functional, persistence
ADM-F-014	Choice locking	4.6	TC-014	Functional, state transition
ADM-F-015	Rank/preference/seat-based allotment engine	4.7	TC-015	Unit, integration, system, oracle
ADM-F-016	Not Allocated determination	4.7	TC-016	Unit, system, oracle
ADM-F-017	Allotment result publication	4.8	TC-017	Functional, state transition
ADM-F-018	Allotment result viewing	4.8	TC-018	Functional, system
ADM-F-019	Seat accept/reject decision	4.8	TC-019	Functional, state transition
ADM-F-020	Admin candidate/application dashboard	4.9	TC-020	Functional, data consistency
ADM-NFR-001	Response time performance	5	TC-026	Performance
ADM-NFR-002	Allotment run scalability	5	TC-027	Performance, scalability
ADM-NFR-003	Seat-count data consistency	5	TC-028	Data consistency, reliability
ADM-NFR-004	Candidate workflow usability	5	TC-029	Usability
ADM-NFR-005	Configurable, data-driven maintainability	5	TC-030	Maintainability, system
SEC-F-001	Mandatory authentication	5.1.2	TC-021	Security
SEC-F-002	Role-based access control	5.1.2	TC-022	Security
SEC-F-003	Salted password hashing	5.1.2	TC-023	Security
SEC-F-004	Encrypted data in transit	5.1.2	TC-024	Security
SEC-F-005	Audit logging of privileged actions	5.1.2	TC-025	Security

13.2 Workflow Acceptance Coverage (SRS Section 6)

Workflow	Covered by
Candidate Registration	TC-001
Application Submission	TC-004, TC-005
Document Upload & Verification	TC-006, TC-007, TC-008
Eligibility Check	TC-009
Choice Filling and Locking	TC-012, TC-013, TC-014
Seat Allotment (core algorithm)	TC-015, TC-016, TC-028
Allotment Result Publication	TC-017, TC-018
Seat Acceptance/Rejection	TC-019
Admin Access Control	TC-022 (and TC-021)

14. Test Metrics & Reporting

Metrics collected:

- % test cases executed (target 100%)
- % passed/failed
- Defect density (defects per module / per requirement)
- Defect aging (days open, by severity)
- Requirement coverage (requirements with at least one executed test case; target 30 of 30)
- Allotment oracle pass/fail and seat-count consistency check result per run

Reports:

- Daily execution status (during execution window)
- Updated RTM status
- Final Test Summary Report

15. Approvals

Role	Name	Signature / Date
QA Lead		
Dev Lead		
Product Owner		

Appendix A: Detailed Test Cases

Priority is taken from the SRS requirement. "Pre" lists preconditions. All data is mock/seeded.

A.1 Candidate Account Management

TC ID	Req. ID	Priority	Test Scenario & Steps	Expected Result
TC-001	ADM-F-001	High	1. Open the registration page and submit valid, unique name, email, mobile, date of birth and password. 2. Register again using the same email. 3. Register again using the same mobile number. 4. Submit with a mandatory field empty / invalid email format.	Step 1: account created and confirmation displayed. Steps 2-3: duplicate email/mobile rejected with a message; no second account. Step 4: rejected with field-level error.
TC-002	ADM-F-002	High	Pre: registered candidate. 1. Log in with email and correct password; repeat with mobile number. 2. Log out. 3. Log in with an incorrect password; with an unregistered email.	Step 1: authenticated and redirected to dashboard. Step 2: session ended; protected pages inaccessible. Step 3: error shown, access denied.
TC-003	ADM-F-003	High	Pre: logged-in candidate; seeded exam records. 1. Edit contact details and address, save. 2. Enter exam type (KCET/COMEDK), rank, category and score matching a seeded record. 3. Enter exam details that match no seeded record.	Step 1: changes saved and shown on the profile page. Step 2: rank and category stored against the candidate. Step 3: not stored as valid; error shown (behaviour to be confirmed, Section 11).

A.2 Application Management

TC ID	Req. ID	Priority	Test Scenario & Steps	Expected Result
TC-004	ADM-F-004	High	Pre: candidate with exam details. 1. Fill the application partially and click Save Draft. 2. Log out, log in again and open the draft.	Draft saved; all entered data present on resume, no data loss.
TC-005	ADM-F-005	High	1. Open a complete draft, edit a field, submit. 2. Open an incomplete draft and submit. 3. Submit a second candidate's application.	Step 1: status becomes Submitted; unique reference number generated and displayed. Step 2: rejected with field-level errors; status stays Draft. Step 3: reference number differs from all earlier ones (non-reusable).

A.3 Document Upload and Verification

TC ID	Req. ID	Priority	Test Scenario & Steps	Expected Result
TC-006	ADM-F-006	High	Pre: submitted application. 1. Upload PDF, JPEG and PNG files within the size limit. 2. Upload a file of an unsupported type (e.g., .docx, .exe). 3. Upload a file just under, at and just over the size limit (2 MB assumed). 4. Rename an .exe to .pdf and upload.	Step 1: upload succeeds. Steps 2-4: rejected with an explanatory message; nothing stored. Boundary: at/under limit accepted, over limit rejected.
TC-007	ADM-F-007	High	Pre: documents uploaded (Pending Verification). 1. Log in as admin; open the verification queue. 2. Approve one document. 3. Reject another with a reason. 4. Try to reject without a reason. 5. Log in as the candidate.	Step 2: status Approved. Step 3: status Rejected; reason stored. Step 4: blocked - reason is mandatory. Step 5: rejection reason visible to the candidate.
TC-008	ADM-F-008	Medium	Pre: one Approved and one Rejected document. 1. As candidate, open the document status view. 2. Upload a replacement for the rejected document.	Step 1: correct status shown for each document. Step 2: replacement accepted; status resets to Pending Verification and appears in the admin queue.

A.4 Eligibility Verification

TC ID	Req. ID	Priority	Test Scenario & Steps	Expected Result
TC-009	ADM-F-009	High	Pre: configured eligibility rules; three seeded candidates. 1. Candidate meeting all rules, documents verified. 2. Candidate failing a specific rule (e.g., category/rank/score rule). 3. Candidate meeting the rules but documents not yet verified. 4. Check eligibility for each.	Case 1: Eligible. Case 2: Not Eligible with a reason naming the rule not satisfied. Case 3: Pending Verification with a reason. Statuses and reasons match the decision table.

A.5 College, Branch and Seat Management

TC ID	Req. ID	Priority	Test Scenario & Steps	Expected Result
TC-010	ADM-F-010	Medium	Pre: seeded colleges, branches, seat matrix. 1. As candidate, open the catalogue. 2. Filter by a college name; then by a branch name. 3. Filter with a term that matches nothing. 4. Compare displayed seat availability with the seat matrix (SQL).	Only matching colleges/branches shown; empty result handled gracefully; seat figures equal the underlying seat matrix.

TC ID	Req. ID	Priority	Test Scenario & Steps	Expected Result
TC-011	ADM-F-011	High	1. As admin, add a college and branch with category-wise seat capacity. 2. Edit the capacity. 3. As candidate, check the catalogue and choice-filling page. 4. Run allotment, then re-check seat counts.	New record immediately visible in the catalogue and available for choices; edited capacity reflected; after allotment, seat counts reflect the number allotted per college-branch-category.

A.6 Choice Filling

TC ID	Req. ID	Priority	Test Scenario & Steps	Expected Result
TC-012	ADM-F-012	High	Pre: eligible candidate (ADM-F-009 passed). 1. Add three college-branch choices. 2. Remove one; preview the list. 3. Re-add a choice already in the list. 4. As an ineligible candidate, open choice filling.	Steps 1-2: list and preview update correctly. Step 3: rejected with a duplicate-choice message. Step 4: choices cannot be added.
TC-013	ADM-F-013	Medium	Pre: list with at least three choices, not locked. 1. Move the third choice to first position. 2. Log out and back in; view the list.	New priority order saved and persisted after re-login.
TC-014	ADM-F-014	High	Pre: ordered, unlocked list. 1. Lock the list. 2. Attempt to add, remove and reorder a choice (UI and direct API).	List locked; every modification attempt rejected; list unchanged and used as-is for allotment.

A.7 Seat Allotment

TC ID	Req. ID	Priority	Test Scenario & Steps	Expected Result
TC-015	ADM-F-015	High	Pre: oracle dataset loaded (Appendix B); all choice lists locked. 1. As admin, run allotment. 2. Compare each candidate's result and the remaining seats with the expected table. 3. Repeat with category-specific seats and an ineligible candidate in the dataset.	Candidates processed in ascending rank; first available eligible preference allotted; seat availability reduces immediately after each allotment; results equal the hand-computed expected values; ineligible candidates receive no seat; category and capacity respected.
TC-016	ADM-F-016	High	Pre: oracle dataset; a candidate whose locked preferences are all full or unavailable. 1. Run allotment. 2. View the result of that candidate.	Candidate recorded as Not Allocated (C4 and C5 in Appendix B).

A.8 Allotment Result and Seat Decision

TC ID	Req. ID	Priority	Test Scenario & Steps	Expected Result
TC-017	ADM-F-017	High	Pre: allotment run completed, results unpublished. 1. As candidate, try to view the result. 2. As admin, publish results. 3. As candidate, view the result again.	Step 1: result not visible. Step 3: result visible immediately after publication.
TC-018	ADM-F-018	High	Pre: results published. 1. Log in as an allotted candidate. 2. Log in as a Not Allocated candidate.	Step 1: correct college/branch and status Allocated. Step 2: status Not Allocated; no college shown.
TC-019	ADM-F-019	High	Pre: published result with an allotted seat. 1. Candidate A selects Accept. 2. Candidate A tries to change the decision. 3. Candidate B selects Reject/Withdraw, then tries again.	Decision status updated accordingly for A and B; the second attempt is rejected - decision can be made exactly once.

A.9 Administration and Audit

TC ID	Req. ID	Priority	Test Scenario & Steps	Expected Result
TC-020	ADM-F-020	Medium	Pre: several candidates with applications and documents in various states. 1. As admin, open the dashboard. 2. Compare candidate counts, application statuses and the pending-verification queue with database counts.	Dashboard counts and queues match the underlying data; records can be viewed and managed.

A.10 Security

TC ID	Req. ID	Priority	Test Scenario & Steps	Expected Result
TC-021	SEC-F-001	High	1. Without logging in, open candidate and admin page URLs directly. 2. Call protected API endpoints without a session token.	Redirected to login; no protected data returned; access denied.
TC-022	SEC-F-002	High	1. Log in as a candidate. 2. Open admin URLs and call admin APIs: edit seat matrix, run allotment, publish results. 3. Log in as admin and perform the same actions.	Step 2: authorization error (e.g., 403); no state change. Step 3: all actions succeed.
TC-023	SEC-F-003	High	1. Inspect the credential store for candidate and admin accounts. 2. Register two users with the same password and compare stored values. 3. Search application logs for passwords.	Only salted hashes stored; same password yields different hashes; no plaintext in store or logs.
TC-024	SEC-F-004	High	1. Capture network traffic while logging in and entering personal data. 2. Access the site over plain HTTP. 3. Check the negotiated TLS version.	Only encrypted traffic observed; no plaintext credentials or personal data; HTTP refused or redirected; TLS 1.2 or higher.

TC ID	Req. ID	Priority	Test Scenario & Steps	Expected Result
TC-025	SEC-F-005	Medium	1. As admin, edit the seat matrix, run allotment and publish results. 2. Open the audit log. 3. Look for any UI/API means to edit or delete entries.	One entry per action with actor, timestamp and action type; entries cannot be edited through the standard UI.

A.11 Non-Functional

TC ID	Req. ID	Priority	Test Scenario & Steps	Expected Result
TC-026	ADM-NFR-001	High	Pre: demonstration environment, seeded data. 1. With JMeter, simulate 25 concurrent candidates (proposed load) performing login, catalogue view and choice filling. 2. Record response times over the run.	Average response time of standard candidate requests ≤ 2 s.
TC-027	ADM-NFR-002	Medium	Pre: full seeded dataset (about 300 candidates, locked choices). 1. Run allotment and time it. 2. Repeat with double the data volume (about 600).	Run completes in under 1 minute for the full demonstration dataset; doubling the data needs no architectural or code changes.
TC-028	ADM-NFR-003	High	Pre: allotment completed (also after repeated runs and seat-matrix edits). 1. For every college-branch-category, count allotment records. 2. Compare with configured capacity and displayed available seats (script/SQL).	Allotted $\leq$ capacity everywhere; available seats = capacity – allotted; no mismatch.
TC-029	ADM-NFR-004	Medium	1. Recruit about five first-time users (no briefing). 2. Ask each to register and submit an application, without help. 3. Record completion, time taken and difficulties.	Majority (at least 3 of 5) complete submission without external assistance.
TC-030	ADM-NFR-005	Medium	1. As admin, add a new college, branch and an eligibility rule using only the admin interface. 2. Verify it appears in the catalogue and is applied in eligibility checks and allotment. 3. Confirm no code change or redeployment was needed.	New data is usable end-to-end through admin actions alone; no source-code changes.

Appendix B: Allotment Oracle Dataset (TC-015, TC-016, TC-028)

A small deterministic dataset with a hand-computed expected result. All candidates are Eligible, category General, with choice lists locked. Higher priority is listed first.

B.1 Seat matrix (before allotment)

College - Branch	Category	Seats
College A - CSE	General	2
College A - ISE	General	1
College B - ECE	General	1

B.2 Candidates and expected result

Candidate	Rank	Locked choices (priority order)	Expected result	Reason
C1	1	A-CSE, A-ISE	A-CSE (Allocated)	First choice free
C2	2	A-CSE, B-ECE	A-CSE (Allocated)	Last A-CSE seat
C3	3	A-CSE, A-ISE	A-ISE (Allocated)	A-CSE full
C4	4	A-CSE, A-ISE	Not Allocated	Both full
C5	5	A-CSE	Not Allocated	Only choice full

Expected seats after the run: A-CSE 0, A-ISE 0, B-ECE 1 (unchanged). Extend the dataset with category-specific seat rows and one Not Eligible candidate whose locked list must be ignored for TC-015 step 3.